

First International Workshop on Emerging Trends in DevOps and Infrastructure

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ABSTRACT

In this paper, we describe the proceedings of first international Workshop on Emerging Trends in DevOps and Infrastructure held at XP-2016. The workshop was conceived and administered to provide a platform for researchers and industrial practitioners researching or working within DevOps environments to come together and discuss issues around the idea of DevOps.

The workshop proceeded with research presentations and discussion on the basics of DevOps such as how to define them to more detailed technical challenges and opportunities of working with DevOps practices within an organizational setting.

Categories and Subject Descriptors

Software creation and management - Software development
process management - Agile software development

General Terms

Algorithms, Management, Measurement, Documentation, Performance, Design, Experimentation, Security, Human Factors, Verification.

Keywords

DevOps, Development, Operations, Emerging trends, Continuous software development, Continuous Delivery, Continuous Experimentation, Decision Support.

1. INTRODUCTION

2. Workshop Theme

The question of how companies like Amazon, Netflix, and Facebook can deliver features regularly to their customers in some cases several times per day while other companies need several months for their releases is a particularly interesting one. According to Duvall [1], the companies listed here have been able to deliver quality software regularly by adopting a “DevOps” approach with or without realizing it. By dismantling the traditional silos and creating an environment where development

and operations teams work together and learn from each other enabled them to meet their business demands.

Recognizing the harmful disconnect that tends to exist between important activities such as planning, development and implementation, Fitzgerald and Stol [2] provide a roadmap for continuous software engineering, which positions a wide variety of “continuous” trends, including continuous integration, continuous delivery/deployment, but also concepts such as continuous trust and continuous usage. The roadmap recognizes that customers do not make a single adoption choice for a specific product, but that customers continuously re-evaluate their decision. For example, incidents may cause issues of trust and security, leading users to potentially abandoning a product in favour of a competitive product. Also, as competing products are continuously updated with new features, customers may expect this from their current product choice also. Thus, software suppliers must also be concerned with the later lifecycle phases of a product. The emerging phenomena such as DevOps, Continuous Delivery, and Infrastructure as Code reflect the attempts to address the disconnects to enable delivery of high quality software in a continuous manner.

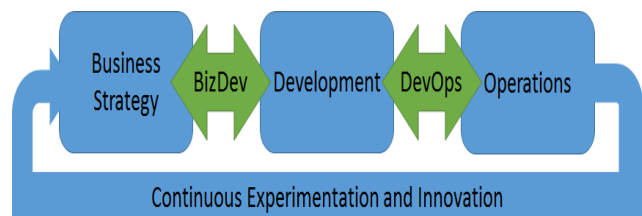


Figure 1: Simplified Continuous* view of activities from business to operations, adapted from [2]

The main theme of proposed workshop will revolve around the four principles suggested by Humble and Molesky [3] that characterize DevOps, namely:

1. **Culture:** DevOps need a cultural shift where code “thrown over the wall” is no longer acceptable, while the responsibility for delivery of high quality software to end users is jointly accepted by the development and operations working together.
2. **Automation:** To achieve shorter lead times, rapid delivery and fast feedback from end users, DevOps relies on full automation for build, testing, and deployment.
3. **Measurement:** The status of current delivery and goal setting for improvement for future deliveries can only

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be done through measurements. DevOps use various forms of measurements and monitoring which include business metrics (e.g. revenue) to metrics for technical overview (e.g. test coverage) and customer satisfaction.

4. **Sharing:** DevOps are built upon the foundation of sharing, the sharing happens at different levels which most often includes sharing of knowledge, tools and infrastructure. Further any failure and successes are also shared which bring the development and operations teams even closer together.

Roche [4] summarized the importance and improvements the DevOps wave has brought in the Quality Assurance domain. He suggests that the specific focus on metrics and process standardization under DevOps principles has significantly improved product development and ownership in many specific areas. DevOps principles have been particularly useful in building large and complex web based projects [5], but how DevOps principles are being used in more traditional industries that are now fast becoming software intensive (e.g. automotive) remains largely unknown.

Elliot's [6] recent survey identifies the high average cost of unplanned application downtime for large organizations. The study also suggests that average cost of critical application failure is between \$500,000 to \$1 million per hour. While the survey show that most respondents are either using or evaluating DevOps practices, some challenges remain impeding their uptake on a larger scale. The three leading challenges that have been consistently reported are:

1. Cultural inhibitors preventing risk taking and teamwork,
2. Fragmented processes within organizations limiting automation and cross-collaboration, and
3. Lack of strong executive stakeholder support.

3. Goals of the Workshop

The DevOps topic is attracting significant attention from both researchers and practitioners -- a Special Issue on the topic will feature in *IEEE Software* in 2016, for example. However, what is needed are more experience reports and "stories from the trenches", to share war stories and identify common trends and patterns. The goal of the workshop is to address some of the above-mentioned knowledge gaps and DevOps-related challenges by gathering practitioners and researchers working on these topics. Specifically the workshop aimed to attract submissions and stimulate discussions around following themes:

- Empirical studies highlighting the need for DevOps in various industrial settings
- Case studies on successful and unsuccessful DevOps experiences from industry
- Challenges/Impediments of migration from traditional development to DevOps oriented development and maintenance
- Metrics for DevOps – including both the technical metrics (e.g. defects), process metrics (e.g. continuous integration) and management (e.g. efficiency)
- Current & future trends relevant to DevOps
- Infrastructure as Code - fast scaling up and down of infrastructure on demand treating the configuration code in the same way as the application code.

4. Papers presented and discussion

The workshop was opened with a keynote delivered by noted software engineering expert Jan Bosch is professor of software engineering and director of the Software Center at Chalmers University Technology in Gothenburg, Sweden. The keynote titled "Speed, Data and Ecosystems: The Future of Software Engineering" provided an interesting overview of the current state of art in software development as well as how speed at which software is developed is most critical factors for organizations delivering software based products and services. The abstract of the talk is provided here:

"We are living in the most exciting time in the history of mankind. The last century has seen unprecedented improvements in the quality of the human condition and technology is at the heart of this progress. Now we are experiencing an even bigger leap as we move towards a new level of digitisation and automation. Ranging from self-driving cars to factories without workers to societal infrastructure, every sensor and actuator is becoming connected and new applications that enable new opportunities are appearing daily. The fuel of this emerging Internet of Things reality is software. There are three areas where the companies seeking to survive and thrive in this new world need to be world leading: speed, data and ecosystems. Speed is concerned with converting new customer insights into deployed solutions in hours and days rather than months and years. Effective use of data coming from the field, both in development as in the applications themselves, is critical to ensure that we are building the right products that can successfully act in a DevOps environment. Finally, successfully integrating ourselves in our business and technology ecosystems such that speed and data-driven development and execution expand beyond the boundaries of the company is crucial for success. The keynote first introduces the aforementioned development, then discusses the key areas of speed, data and ecosystems and finally presents the implications for organisations that seek to continue to be successful."

The first paper presented at the workshop was by Jabbari et al. "What is DevOps? A Systematic Mapping Study on Definitions and Practices", the author's systematically mapped 44 studies published on DevOps to answer a basic question: What is DevOps? By analysing and comparing the definitions used in research literature of DevOps, authors identified and classified the practices associated with what is referred as DevOps. The study investigated and seeks to answer following questions:

1. How is DevOps defined in peer-reviewed literature on the topic?
2. Which practices were associated with DevOps in the literature?
3. What are the similarities and differences reported by the authors of primary studies between DevOps and the other development methods?

Based on the observations from literature, the authors proposed the following definition for DevOps:

"DevOps is a development methodology aimed at bridging the gap between Development (Dev) and Operations , emphasizing communication and collaboration, continuous integration, quality assurance and delivery with automated deployment utilizing a set of development practices."

The second paper presented at the workshop was presented by Pauli Kärpänoja of Solita, Inc. based in Finland. The paper

“Exploring Peopleware in Continuous Delivery” explored the perspective of developers where continuous delivery practices are applied. The authors also attempt to explain the prerequisites for successful introduction of continuous delivery framework in an organizational unit, specifically following research questions are attempted to be answered in the research presented:

1. How do continuous delivery practices affect software developers?
2. What are the prerequisites for continuous delivery?

According to authors research results presented at workshop, introduction of continuous delivery practices improves software developer’s team motivation and commitment to their organization. The improvement in team morale and cohesiveness also impacts positively the quality of product delivered and stability of deployments. Further the authors observed that these improvements do not come at cost of stress for software developers nor does it need extra workhours.

With respect to prerequisites for successfully introducing the continuous delivery practices in an organization, the most critical factor identified was the skillset of team where members are ready to take the responsibility. The desire to commit and follow through the agreed procedures was also deemed important prerequisite for smooth introduction of continuous delivery practices.

The further two papers in the workshop were focused on the technical side of DevOps. Federico Giaimo from University of Gothenburg in Sweden presented their research titled “Continuous Experimentation on Cyber-Physical Systems: Challenges and Opportunities”. With cyber-physical systems encompassing the interconnected embedded systems, and Internet-of-Things (IoT) taking an ever increasing role in software products and services we interact with, the author claim that continuous experimentation is under-explored in this context. Also in many cases continuous experimentation is not considered due to safety considerations. The research paper presented highlighted challenges and opportunities of continuous experimentation for cyber-physical systems. Hardware constraints and evolution, need for feedback data from humans and other systems, requirement of safety guarantees, and involvement of larger and diverse stakeholders were some of the important challenged discussed in their research.

The final paper presented at the workshop by Riccardo Scandariato titled “Traceability Data in Early Development Phases as an Enabler for Decision Support” introduced their method for building decision support system based on rich historical traceability data. The authors reported their vision to generate early warnings regarding problems due to problematic requirements, ill-designed

architectural components, critical deployments and so on. Being able to detect and visualize the patterns indicating early warning signs to relevant stakeholders provides opportunities for early intervention thus enabling better product design and delivery of better quality software with shorter lead time.

5. Conclusion

The first International Workshop on Emerging Trends in DevOps and Infrastructure attracted research and participants that covered the four principles suggested by Humble and Molesky [3] that characterizing DevOps: Culture, Automation, Measurement, and Sharing. The presentations and participants span from academia to industry. The research presented and discussions also ranged from what is DevOps, how it could be defined to detailed technical concepts to improve continuous experimentation and development of decision support systems to enable continuous delivery of high quality software.

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